DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03D: MARTIAL ARTS & TECHNIQUES (PART 4 — QINGGONG, SWORD FLIGHT, CELESTIAL LEVITATION, STEALTH/RECON ARTS & COMBAT MOBILITY)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Context**:
  - `03A`: Combat Moves (`COMBAT_TECHNIQUE` / `moveDefs`), `SK.list` Cache, `chooseSkill` AI, 16 Weapon Mismatches & Spatial Range.
  - `03B`: Composable Engine, 34 Elemental Reactions (`DARK_YIN` vs `YIN_SHADOW`), Passive Loadout & Duet Synergies.
  - `03C`: Mantras & Inner Qi Circulation (`MANTRA` / `MANTRA_SYSTEM`), Lifespan Traps & Sect Formation Warfare.
  - **`03D` (This Document)**: Exhaustive audit of **Movement, Qinggong, Aerial Flight, Stealth & Reconnaissance Arts (`LIGHTNESS_ART`, `MOVEMENT_TECHNIQUE`, `role: 'MOVEMENT_STEALTH_RECON'`, `CULT.QINGGONG`)** across Sect Map pathfinding, Combat positioning/evasion, World Map expeditions, Intelligence/Scouting, and Autonomous NPC decision-making.

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0. Core Architectural Invariants to Uphold

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order**: Preserve exact `<script>` load sequence in `index.html` and `test/load.js`.
3. **100% Emergent NPC Autonomy**: NPCs must autonomously decide **when to learn** Qinggong/Recon arts based on their daily duties (scouting, outer-peak commuting, herb gathering, combat kiting) and **which locomotion mode to use** given weather, distance, path fantasy, and Qi/stamina reserves—never relying on manual UI toggles.
4. **Zero-RNG Read Purity & Tick Budget**: `Render.draw` and pure stat/UI readers must never mutate world state or advance `W.rng`; maintain 80-NPC `SIM.tick <= 3.5 ms/tick`.

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1. [QG-CAT-01] Catalog Misclassification & Fake Stealth/Recon Mechanics Across 38+ Techniques

1.1 Half of the ART190 Footwork & Marching Arts Are Excluded From CULT.QINGGONG (Misrouted Into Healing/Support Auras!)

- **Locations**: data/data_tech.js (ART190_XIAN_041..052, ART190_BODY_041..052, ART190_GOD_041..052), js/cultivation.js (CULT.QINGGONG), test/v540_qinggong_movement_and_flight_suite.js, test/v570_support_and_auxiliary_arts_suite.js
- **Mechanism of Failure**:
  - In data_tech.js, the 36 role: "MOVEMENT_STEALTH_RECON" arts (041..052 per path) alternate cat between "LIGHTNESS_ART" (odd numbers) and "AUXILIARY_ART" (even numbers).
  - Patch v5.4.0 enriched **only** cat === 'LIGHTNESS_ART' || cat === 'MOVEMENT_TECHNIQUE' with qinggongMode / qinggongSpec. Then Patch v5.7.0 (v570 line 69) swept **all** cat === 'AUXILIARY_ART' into CULT.SUPPORT_ARTS!
  - **Absurd Runtime Result**:
    1. **True Footwork & Water-Walking Manuals Give Zero Qinggong Movement**:
       - ART190_BODY_050 (ก้าวเหยียบคลื่นสมุทรไร้จม — Tier-6 Ocean-Wave Walking: "ใช้พละกำลังขาวิ่งบนผิวน้ำ ข้ามแม่น้ำและบึงพิษรวดเร็ว ประหยัดเสบียงเดินทาง") has cat: "AUXILIARY_ART", so learning it **does NOT unlock WATER_LOTUS_DRIFT, gives 0% walk speed, and gives 0% river-route expedition bonus**!
       - ART190_BODY_042 (ก้าวทะยานมังกรเหยียบผา — Cliff-Stepping Dragon Leap), ART190_BODY_048 (ท่าร่างกบหยกดีดตัวหลบคม — Jade Toad Bound), ART190_BODY_052 (เคล็ดวิ่งผลัดพันลี้ไม่สิ้นแรง — Thousand-Li Sprint), ART190_XIAN_048 (ก้าวเหยียบน้ำค้างไร้เสียง), ART190_GOD_042 (ก้าวเหินตามแสงดาวพยากรณ์), and ART190_GOD_052 (วิชาเดินทัพใต้ร่มเงาเทวรูป) are **all excluded from CULT.QINGGONG**!
    2. **Sensory Eye/Ear Arts Falsely Grant Running Speed**:
       - Conversely, odd-numbered sensory arts tagged cat: "LIGHTNESS_ART"—such as ART190_XIAN_043 (เนตรทิพย์ส่องชีพจรฟ้าดิน — Clairvoyant Eye), ART190_XIAN_047 (วิชาสดับเสียงลมพันลี้ — Wind-Listening Ear), ART190_BODY_043 (วิชาฟังเสียงชีพจรดิน — Ear-to-Ground Tremorsense), and ART190_BODY_051

(วิชาคำรามเปิดโปงเงามายา — Mirage-Shattering Roar)—are mapped into running/leaping Qinggong modes even though they are **perception/detection arts**, not locomotion!

1.2 100% Unimplemented Stealth, Scouting, Tracking & Counter-Espionage Promises

- **Verified Fact**: Searching the entire repository for `'MOVEMENT_STEALTH_RECON'` matches **only `data_tech.js`, `data_martial.js`, and `skills.js` (`arch = 'evade'`)**—zero hooks exist in `explore.js`, `expedition.js`, `sect.js`, or `wild_resources.js`!
- **Dead Lore Promises Across 20+ Manuals**:
  - **Stealth & Scent-Masking** (`ART190_XIAN_042` เร้นเงาหมอกคราม, `ART190_XIAN_044` ลบกลิ่นอายตัดรอยปราณ, `ART190_BODY_047` กลบกลิ่นเลือดกลางป่า, `ART190_GOD_043` ควันธูปเร้นกาย): Promise avoiding wild beast aggro, slipping past stronger patrols, and escaping pursuit—**none of these reduce encounter/ambush rates or boost flee chance**!
  - **Clairvoyance, Trap-Sense & Fortune Discovery** (`ART190_XIAN_043` เนตรทิพย์, `ART190_BODY_043` ฟังเสียงชีพจรดิน, `ART190_GOD_041` เนตรพยากรณ์, `ART190_GOD_051` ฟังเสียงวิญญาณพงไพร): Promise trap evasion, ambush pre-warning (`first-strike dodge`), and finding hidden herbs/fortune nodes—**completely unwired in `EXPLORE` and `COMBAT`**!
  - **Tracking & Pursuit** (`ART190_XIAN_049` ประทับตราดาราสะกดรอย, `ART190_BODY_049` สะกดรอยด้วยกลิ่นโลหิต, `ART190_GOD_050` ประทับตราบาปติดตามผู้รุกราน): Promise tracking fleeing enemies, defectors (`resolveDefectorUltimatum`), and clue trails (`visitors_clues.js`)—**completely unwired**!
  - **Camp Counter-Espionage Fog** (`ART190_XIAN_050` ม่านหมอกพรางค่ายรบกวนข่าว, `ART190_GOD_048` พรางค่ายด้วยมิติควันธูป): Promise blocking enemy spies from reading sect strength—**completely ignored by `SECT` intelligence/raid targeting**!

1.3 Remediation Blueprint (`[QG-CAT-01]`)

- Classify `role === 'MOVEMENT_STEALTH_RECON'` arts by semantic sub-tag (`locomotion`, `stealth_escape`, `sensory_recon`, `pursuit_track`, `camp_counter_intel`) **without changing `JD.MARTIAL.techniques.length` or breaking `v570`'s `supportArchetype` property**:
  1. Attach `qinggongMode` / `qinggongSpec` to all true locomotion `AUXILIARY_ART` manuals (`ART190_BODY_042 -> LEAP_BOUND`, `048 -> LEAP_BOUND`, `050 -> WATER_LOTUS_DRIFT`, `052 -> SWIFT_STRIDE`, `ART190_XIAN_048 -> WATER_LOTUS_DRIFT`, `ART190_GOD_042 -> SHADOW_BLINK`, `052 -> SWIFT_STRIDE`).

2. Add a fast `CULT.RECON_STEALTH.evaluatePartyRecon(party)` helper (`O(party_size)`) wired into:
    - **`explore.js` & `expedition.js`**: `sensory_recon` reduces trap/ambush hazard by up to `-28%` and boosts hidden fortune/herb discovery by `+15%`; `stealth_escape` boosts retreat/flee success (`fleeThreshold` / `EXPED` disengage) by `+22%` and reduces hostile sect encounter clash rate when outnumbered (`powerA < powerB * 0.85`)!
    - **`visitors_clues.js` & `sect.js` (`intel` / `scout` / `warden`)**: `pursuit_track` boosts clue-gathering progress and defector interception; `camp_counter_intel` learned by the `intel` or `arraymaster` officer masks sect treasury/power from rival sect raid opportunism and adds fog-of-war defense!
    - **`combat.js`**: `sensory_recon` (เนตรทิพย์ / คำรามเปิดโปงเงามายา) counters enemy `shadowClone` / `shadowVeil` evasion (`+15 acc` vs concealed/clone targets)!

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2. `[QG-RESOURCE-02]` Zero-Qi (`SHADOW_SAMURAI`) & Pure Flesh-Body (`BODY` / `ARENA_GLORY`) Flight Paradox

2.1 Hard `qi < minQiPct` Gate vs Lore-Breaking Qi Sword/Cloud Flight

- **Locations**: `js/cultivation.js` (`CULT.QINGGONG.MODES`, `resolveNpcMode`, `stepQinggongMove`)
- **Two-Sided Logic Defect**:
    1. **Zero-Qi Lockout**: Every Qinggong mode above `MORTAL_WALK` defines `qiCostPerTick` (`0.15` to `0.75`) and `minQiPct` (`10%` to `20%`). For **`SHADOW_SAMURAI`** (`CULT.qiMax(n) === 0`, `n.h.state.qi === 0`), checking `qi < qiMax * minQiPct || qi < qiCostPerTick` either permanently triggers `exhaustedFallback -> 'MORTAL_WALK'` (making `TECH_V230_SHADOW_SAMURAI_QINGGONG` useless on the map!), OR if bypassed for free, lets Zero-Qi Samurai fly indefinitely for `0` resource cost!
    2. **Thematic & Visual Contradiction (`render.js`)**: A `SHADOW_SAMURAI` (mortal zero-Qi swordsman), `NINPO_SHADOW` shinobi, `ARENA_GLORY` pit fighter, or pure muscle `BODY` / `DEVOURING_FLESH` / `PRIMORDIAL_SHIFTER` cultivator **standing atop a glowing Xianxia Flying Sword (`SWORD_FLIGHT`) or floating on auspicious Taoist clouds wrapped in celestial silk ribbons (`CELESTIAL_LEVITATION`)** violates path identity!
- **Remediation Blueprint**:
    - **Resource Substitution**: For `CULT.isZeroQiSamurai(n)` (and stamina-dominant paths when `qiMax` is minimal), pay locomotion upkeep from

Stamina (`h.state.energy`) at `energyCost = qiCostPerTick * 0.08`, falling back to `MORTAL_WALK` when `energy < 22` and resuming at `energy >= 38`.

- **Path-Authentic High-Tier Locomotion Mapping**:
  - `SHADOW_SAMURAI` & `NINPO_SHADOW`: Cap/map high-tier locomotion to **`SHADOW_BLINK` (`ก้าวพริบตาไร้เงา / ชุนโป`)** enhanced at Stage 6+ to `×2.65` speed with zero cloud/flying-sword visuals!
  - `BODY`, `ARENA_GLORY`, `DEVOURING_FLESH`, `PRIMORDIAL_SHIFTER`: Map high-tier locomotion (unless explicitly holding a `Kunpeng` / `Flight` scripture) to **Titanic Sky-Bounding (`LEAP_BOUND` Tier-2 Overdrive: `×2.55` speed, 4-tile vaulting arcs)** rather than Taoist cloud ribbons!

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3. `[QG-MAP-03]` Mid-Flight Qi Exhaustion Over `MAP.blocked` Terrain Traps NPCs Inside Cliffs, Lakes & Roofs!

3.1 Exact Pathfinding & Collision Bug (`js/ai.js` — `AI.buildAerialPath` & `AI.stepMove`)

- **Locations**: `js/ai.js` (`AI.goTo`, `AI.buildAerialPath`, `AI.stepMove`), `js/combat.js` (`COMBAT.enter`, `checkLineOfSight`)
- **Step-by-Step Reproduction**:
  1. An NPC with `SWORD_FLIGHT` or `CELESTIAL_LEVITATION` calls `AI.goTo(W, n, tx, ty)` over a distance ≥ 3.5 tiles.
  2. `AI.buildAerialPath` constructs a direct straight-line trajectory (`aerialFlightActive = true`) passing directly over **`MAP.blocked(W.map, x, y) === true` tiles** (deep water lakes, impassable mountain cliffs, or solid building footprints).
  3. **Mid-flight**, while `(n.x, n.y)` is directly over a blocked lake/cliff/roof tile:
     - **Case A (Qi Exhaustion)**: `stepQinggongMove` drains Qi below `20%` (`SWORD_FLIGHT` in storm drains `1.00 Qi/tick`!). `resolveNpcMode` flips `exhaustedFallback = true` (`modeId = 'MORTAL_WALK'`), sets `n.aerialFlightActive = false`, and attempts ground A* replanning **starting from the blocked tile `(n.x, n.y)`**!
     - **Case B (Sudden Raid / Combat Interrupt)**: `COMBAT.enter(W, f, n.id)` sets `n.path = null` while `n` is hovering on a `MAP.blocked` tile!
  4. **Catastrophic Result**:
     - Ground A* from an enclosed blocked tile fails or traps the NPC inside walls/deep water where they can never reach food or a meditation cushion, eventually starving!

    - In combat, standing on a `MAP.blocked` tile makes melee units unable to walk adjacent to `n` and causes `COMBAT.checkLineOfSight` (`MAP.blocked` raycast) to permanently block ranged shots!
- **Why `v540` Missed It**: `v540_qinggong_movement_and_flight_suite.js:120–129` only stepped **1 tick** from a walkable starting tile before setting `qi = 5%`, so `(elder.x, elder.y)` happened to still be on walkable ground!
- **Remediation Blueprint**:
    - Implement `MAP.nearestWalkableTile(map, x, y, maxR = 14)` (expanding ring search).
    - Whenever an NPC transitions from `aerialFlightActive === true` to grounded (due to `exhaustedFallback`, `COMBAT.enter`, `AI.interruptFor`, `grounded` debuff, or destination arrival) while `MAP.blocked(W.map, n.x, n.y)` is `true`, **immediately glide-land `(n.x, n.y)` onto `MAP.nearestWalkableTile`** (`n.fx = { t: W.t, text: '🪶 ร่อนลงสู่พื้น' }`) *before* clearing `aerialFlightActive` or running ground A*!

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4. `[QG-CMB-04]` Combat Mobility Split-Brain: NPCs Ignore `slow`/`haste`/ZoC & Early `return` Breaks Kiting/Blink!

4.1 `COMBAT.moveToward` Applies `ST.get(u).move` and `zocSlowMult` ONLY to `u.foe` — Sect NPCs Ignore Both!

- **Location**: `js/combat.js:211–256` (`moveToward`)
- **Exact Code**:

```JavaScript
// Lines 212-230 compute zocSlowMult = 0.52 ...
if (u.foe) {
  const sp = (u.speed || 0.6) * (run ? 1.0 : 0.8) * clamp(ST.get(u).move / 100, 0.35, 2.0) * zocSlowMult;
  ...
} else {
  const gx = Math.floor(tx), gy = Math.floor(ty); ... AI.goTo(W, u, gx, gy);
  AI.stepMove(W, u);
}
```

- **Three Simultaneous Bugs in Those 15 Lines**:
    1. **NPC Immunity to Combat Movement Stats & ZoC Slow**: In the `else` branch (`!u.foe` — all Sect Disciples & Elders in raids, grudge duels, tournaments, and

leadership duels!), `AI.stepMove` uses `AI.walkSpeed(W, u)`, which **never multiplies by `clamp(ST.get(u).move / 100, 0.35, 2.0)` and never multiplies by `zocSlowMult`**! Thus `slow`, `chill`, `haste`, `THUNDER_STEP_STANCE` (`+move`), subpath `f.move`, and enemy Tank Zone-of-Control (`0.52x` slow) have **0% effect on NPC movement speed**!

2. **Combat Flight Burns Out Skill Qi Pool**: Calling `AI.goTo` + `AI.stepMove` during `fightStep` triggers map-level `SWORD_FLIGHT` (`aerialFlightActive = true` when `dist >= 3.5`), burning `0.75–1.00 Qi/tick` while flying straight over frontline spear ZoC without anti-air interaction!
3. **Orphaned `'grounded'` Status (`combat.js:203`)**: `moveToward` checks `ST.has(u, 'grounded', tNow)`, yet **zero techniques in the entire game ever apply `'grounded'`**—not even `ART190_BODY_039` (เพลงหอกยาวมังกรเงินแทงเมฆ — "ฉุดรั้งผู้ใช้วิชาตัวเบาลงสู่พื้น") or `ART190_GOD_032` (วิชาหอกสายฟ้าพญาครุฑ — "ฉุดกระชากผู้เหาะเหินลงมา")!

4.2 Early `return` in `COMBAT.useSkill` (`lines 886–930`) Completely Skips Retreat Leap-Back, Pure Blink Teleport, & `onCast` Procs!

- **Location**: `js/combat.js:886–966` (`useSkill`)
- **Step-by-Step Proof**:
  1. In `chooseSkill` (`line 705`), when a `RANGED_KITE` archer/zither user is cornered at point-blank (`curDist <= 1.35`), AI boosts `isRetreat` (`arch === 'evade'`) moves by **`+2.10`** so the archer can leap backward out of melee!
  2. `useSkill` executes line 886:

```JavaScript
if (p.arch === 'evade') {
  ST.addStatus(a, 'evadeUp', 14, null, W.t);
  if (p.grantBuff && ST.applyNamedBuff) ST.applyNamedBuff(a, p.grantBuff, buffDur,
W.t, { casterId: a.id });
  if (d) { d._closedOnBy = a.id; a._gapClosedOn = d.id; }
  return; // <--- EARLY RETURN AT LINE 890!!
}
```

  3. **Everything Below Line 890 Is Skipped**:
     - Line 892 (`runPassiveProcs(W, f, a, d, 'onCast', 0)`) is **skipped** (also skipped for `guard`, `shield`, and `counter`!).
     - Line 961 (`if (mSpec.isRetreat) { ... projectSafePosition(+1.6 tiles away); aTac.closedOnBy = null; }`) is **100% UNREACHABLE DEAD CODE** for `arch === 'evade'`! Worse, line 889 sets `d._closedOnBy = a.id` instead of clearing `aTac.closedOnBy = null`, leaving the archer pinned at `dist = 1.0`!

    - Line 929 (`if (p.arch === 'blink') { ... if ((p.M || 0) <= 0) return; }`) **also returns early before line 952**, so non-damaging spatial `blink` skills **never change `(a.x, a.y)`**!
4. **Passive `hasLightness` Turn-1 Free Gap-Close (`combat.js:306`)**: Merely owning any lightness art in `c.techs` (`pa.hasLightness === true`) permanently triggers `DASH_CLOSE_COUNTER` (`+10% dmg, +8 acc`, setting `dTac.closedOnBy = a.id`) on **every basic attack from turn 1**, even without casting a lightness move or while `rooted`/`grounded`!

4.3 Remediation Blueprint (`[QG-CMB-04]`)

- In `AI.walkSpeed(W, n)`: when `COMBAT.isUnitInActiveFight(W, n)`, multiply speed by `clamp(ST.get(n).move / 100, 0.35, 2.0) * (n._zocSlowMult || 1.0)` and cap map-level `SWORD_FLIGHT` to tactical combat stride so fighters don't drain their skill Qi pool walking 2 tiles.
- Move the `Gap-closer` (`isGapCloser`) and `Disengage / Retreat` (`isRetreat`) position displacement block + `runPassiveProcs(..., 'onCast')` **above** the `evade` / `blink` early returns in `useSkill`! For `RANGED_KITE` casting `evade` (`isRetreat`), leap `1.6` tiles **away** and clear `aTac.closedOnBy = null`; for melee fighters casting `evade` when `curDist > 1.8`, dash **in** and set `dTac.closedOnBy = a.id`.
- Require `ST.has(a, 'evadeUp') || ST.hasBuff(a, 'astralMirage') || isGapCloserMove` (and `!ST.has(a, 'root') && !ST.has(a, 'grounded')`) for `DASH_CLOSE_COUNTER` in `resolveWeaponTactics`, rather than granting it passively forever just for owning a book!
- Wire anti-air skills (`ART190_BODY_039`, `ART190_GOD_032`, `GRAVITY_DAO` field) to inflict `'grounded'` (`12` ticks, stripping `evadeUp` / `aerialFlightActive`).

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5. `[QG-EXPED-05]` World-Map Expedition Escort Physics, `useQinggong` Defaults & Spirit-Vessel Arts

5.1 Escort Speed Physics & Unwired Spirit-Boat Manuals (`expedition.js`)

- **Locations**: `js/expedition.js` (`EXPED.createParty`, `EXPED.estimateTravelTime`), `js/cultivation.js` (`CULT.QINGGONG.evaluatePartyExpeditionBonus`), `data/data_tech.js` (`ART190_XIAN_071`, `ART190_GOD_071`)
- **Logical & Contextual Issues**:

1. **Mixed-Realm Party Bottleneck**: If 1 Stage-6 Elder (`CELESTIAL_LEVITATION`) travels with 4 Stage-1 novices (`MORTAL_WALK`), a party cannot march at `×2.90` immortal cloud speed unless the high-realm flyer **carries** junior disciples on their sword/cloud aura (realistic capacity: 1 flyer can carry up to `1` junior on `SWORD_FLIGHT` or `2` juniors on `CELESTIAL_LEVITATION`, or more if someone knows Spirit-Vessel navigation `ART190_XIAN_071` เคล็ดจัดสรรปราณเดินเรือเหินเวหา / `ART190_GOD_071` เคล็ดคำนวณฤกษ์ยามเดินนาวาปราณ!). Otherwise, party march speed must blend toward the slowest uncarried member!
2. **Autonomous `useQinggong` Evaluation**: Ensure `EXPED.createParty` defaults `useQinggong` to `true` (unless party average energy/Qi is critically low or stealth march is chosen) across **all** autonomous expedition callers (`explore.js`, `sect.js` convoys, vassal relief, node claims), and deducts a modest arrival stamina/Qi cost (`-12%` Qi on `SWORD_FLIGHT` vs `-4%` on `CELESTIAL_LEVITATION`) when rushing into immediate combat!

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6. `[QG-AI-06]` NPC Brain (`COG`) Under-Scores Qinggong Manuals (`+0` Category Weight) & Never Adapts Flight Mode

6.1 Why Scouts, Envoys & Outer-Peak Disciples Rarely Study Qinggong (`cognition.js:322–355`)

- **Location**: `js/cognition.js` (`COG.techPreference`)
- **Exact Defect**:
  - `COG.techPreference` awards `+16` to Passives and `+14..26` to Combat Techniques, while checking `if (t.role === 'MOVEMENT_STEALTH_RECON') s += 6 + agility / 10`.
  - Canonical Qinggong manuals (`LIGHT_SWALLOW_STEP`, `LIGHT_CLOUD_STEP`, `LIGHT_THUNDER_SHRINK_EARTH`, `QG_SWIFT_GALE_STRIDE_T1` .. `QG_IMMORTAL_VOID_LEVITATION_T7`, and `MOVEMENT_TECHNIQUE` entries) **do not set `role: 'MOVEMENT_STEALTH_RECON'`** (`role` is `undefined`!), so they receive **`+0` category score** and sink to the very bottom of every NPC's study queue!
  - Furthermore, an NPC who **owns zero movement arts** gets no "missing essential slot" urgency bonus—even if they are appointed **`scout`**, **`envoy`**, **`warden`**, **`quartermaster`**, **or live on a distant Peak (`n.peakId`)**!
- **Remediation Blueprint**:
  1. In `COG.techPreference(n, t)`: recognize all Qinggong/Movement/Recon arts (`t.qinggongMode || t.cat === 'LIGHTNESS_ART' || t.cat === 'MOVEMENT_TECHNIQUE' || t.role === 'MOVEMENT_STEALTH_RECON'`).

    - If `n` currently knows **zero** Qinggong arts, add **`+18` ("ยังไร้พื้นฐานวิชาตัวเบา")**!
    - If `n` knows a lower-tier Qinggong mode than a library manual would unlock (e.g., Stage 4 Sword cultivator unlocking `SWORD_FLIGHT` or Stage 6 Elder unlocking `CELESTIAL_LEVITATION`), add **`+15` ("ยกระดับขอบเขตเหินเวหา")**!
    - Add role weights (`+10` for `post` in `['scout', 'envoy', 'intel', 'vanguard', 'warden']`, `+6` if `n.peakId` or `aspiration === 'wanderer'` / `'fortune_seeker'`).
2. **Contextual Mode Selection (`resolveNpcMode` when `preferredMode === 'AUTO'`)**:
    - During `rain`/`storm`/`thunder_tempest`, if an NPC has both `SWORD_FLIGHT` (`1.00 Qi/tick` + `-15%` turbulence + energy drain) and `WATER_LOTUS_DRIFT` (`0.22 Qi/tick`, `×2.32` over wet/water tiles, `0` storm penalty), smart/cautious NPCs (`rationality >= 58` or `qi < 55%`) should dynamically prefer `WATER_LOTUS_DRIFT` / `SHADOW_BLINK`!
    - During `W.threat.alert` or `W.threat.raid` before combat starts, conserve Qi (`cap auto-flight if qi < 65%`) so defenders don't enter a Sect Raid at `21%` Qi from flying around doing chores!

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7. `[QG-MASTERY-07]` Flying, Leaping & Travelling Grant Zero Mastery (`c.tm`) or `MOVEMENT` Skill Growth

- **Locations**: `js/cultivation.js` (`stepQinggongMove`), `js/expedition.js` (`EXPED.complete`)
- **Defect**: An NPC can fly `10,000` tiles (`totalTilesFlown`), vault `500` bounds (`totalLeaps`), and complete `30` expeditions while their equipped lightness manual (`LEG_LIGHT_LINGBO_STEP`, `QG_IMPERIAL_SWORD_FLIGHT_T5`, etc.) remains stuck at **`c.tm[techId] === 0`** and `h.skills.MOVEMENT` barely grows!
- **Fix**:
    - Every `25` tiles traveled via active Qinggong (`stepQinggongMove`) and upon completing any World Map expedition (`+1.5` mastery), grant `SK.gainMastery(W, n, activeQinggongTechId, 0.15)` and `MARTIAL.skillPractice(n, 'MOVEMENT', 1/60, 1.0)`.
    - Scale Qinggong speed (`+0%` to `+12%`) and Qi cost reduction (`-0%` to `-18%`) by `clamp(c.tm[activeQinggongTechId] || 0, 0, 100) / 100` so veteran flyers and scouts visibly outpace novices using the same manual!

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8. Implementation & Verification Checklist for GPT / Gemini (03D)

- [ ] **Catalog & Recon Wiring**: Add `qinggongMode` to locomotion `AUXILIARY_ART` entries (`ART190_BODY_042/048/050/052`, `ART190_XIAN_048`, `ART190_GOD_042/052`); implement `CULT.RECON_STEALTH.evaluatePartyRecon` hooks across `explore.js` (trap/ambush/discovery), `expedition.js` (stealth disengage & spirit-boat capacity), `sect.js` (counter-espionage fog), and `combat.js` (anti-stealth/clone accuracy).
- [ ] **Safe Aerial Landing**: Add `MAP.nearestWalkableTile` snap whenever `aerialFlightActive` terminates over `MAP.blocked` terrain (test specifically by draining Qi below `20%` while mid-flight over water/cliff tiles!).
- [ ] **Zero-Qi & Body Fantasy**: Use Stamina (`energy`) instead of Qi for `SHADOW_SAMURAI` Qinggong, and map `SHADOW_SAMURAI` / `NINPO_SHADOW` / pure `BODY` high-tier movement to `SHADOW_BLINK` / Titanic `LEAP_BOUND` instead of Flying Swords / Immortal Cloud Ribbons.
- [ ] **Combat Mobility Fixes**: Apply `ST.get(n).move` and `zocSlowMult` to `!u.foe` NPCs during active combat; fix `COMBAT.useSkill` early `return` on `evade` / `blink` so `isRetreat` leaps backward, clears `closedOnBy`, fires `onCast`, and teleports pure `blink`; require active lightness buff/move for `DASH_CLOSE_COUNTER`; wire `'grounded'` status onto anti-air skills.
- [ ] **AI Study & Mastery**: Boost missing-Qinggong priority in `COG.techPreference`, add weather/alert Qi conservation in `'AUTO'` mode, enforce escort carry physics in `evaluatePartyExpeditionBonus`, and award `c.tm` + `MOVEMENT` XP during real Qinggong travel.